Password Auditing and Hash Recovery Assessment Report

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Environment	Controlled Kali Linux lab environment
Primary tools	Ophcrack, Johnny, John the Ripper, Hash Identifier, Hashcat
Assessment focus	Password auditing, hash identification, and recovery workflow analysis

Executive Summary

This report presents the results of a controlled password auditing and hash recovery assessment performed in a Kali Linux lab environment. The assessment evaluated the use of Ophcrack, Johnny, John the Ripper, a hash identification utility, and Hashcat against Windows SAM-derived password data and Unix/Linux shadow-based hashes.

The objective was to compare tool effectiveness, document operational constraints, and evaluate how file format, hash type, and workflow preparation affect recovery outcomes. The assessment demonstrated that different tools produced different recovery rates depending on the input source and algorithm compatibility.

Observed results included recovery of 5 of 12 passwords with Ophcrack, 7 recovered passwords using Johnny and John the Ripper against SAMPWdump-derived input, 6 recovered passwords using John the Ripper against shadow-file data, and 6 cracked hashes with Hashcat after input formatting and algorithm-selection issues were resolved.

Introduction

This assessment was conducted to document password auditing and hash recovery techniques in a controlled environment and to evaluate how common password recovery tools perform across multiple credential sources. The testing focused on Windows LM and NTLM hash data, Unix/Linux shadow-file content, and the preparation required to support successful recovery attempts.

In addition to comparing recovery results, the assessment captured the practical issues that affect workflow success, including incompatible source files, incorrect hash formatting, and algorithm mismatches. These factors are important because password recovery effectiveness is influenced not only by the tool itself, but also by the quality and structure of the input being processed.

Scope

The scope of this assessment was limited to authorized lab-based password auditing activity using previously collected hash material. The test included Windows password recovery attempts using Ophcrack and John the Ripper, Unix/Linux shadow-file analysis with John the Ripper and hash-identification tooling, and Hashcat-based recovery after algorithm and file-format validation.

Primary tools used during the assessment were Ophcrack, Johnny, John the Ripper, Hash Identifier, and Hashcat.

Methodology

The assessment followed a staged workflow designed to compare tool compatibility and recovery performance across several types of hash input. Initial testing used Ophcrack to evaluate Windows-oriented recovery against LM and NTLM data. Follow-on testing used Johnny and John the Ripper to compare GUI-based and command-line approaches, including recovery against both SAMPWdump-derived files and shadow-file content.

A hash identification utility was then used to determine the likely algorithm type of selected hashes before starting Hashcat. Final testing with Hashcat required troubleshooting related to algorithm selection and shadow-file formatting. Once those issues were corrected, recovery was rerun and the resulting cracked hashes were reviewed.

Assessment Snapshot

Tool	Input Source	Recovered	Notes
Ophcrack	Windows SAM-derived hashes	5 of 12	Worked after switching from shadow-file input to SAMdump2-compatible data.
Johnny	Windows hashes	7	GUI-based recovery; less flexible than John the Ripper for shadow-file workflows.
John the Ripper	SAMPWdump-derived file	7	Recovered passwords from Windows-based hash input.
John the Ripper	Shadow file	6	Recovered passwords from Unix/Linux-style shadow-file hashes.
Hash Identifier	Selected hashes	N/A	Used to validate likely hash type before Hashcat execution.
Hashcat	Prepared crack1.hash file	6	Succeeded after correcting algorithm selection and file formatting issues.

Findings and Analysis

1. Ophcrack Results

Ophcrack was used first to evaluate Windows password recovery against LM and NTLM hash material. It is an open-source password recovery tool that uses rainbow tables and supports several import formats, including hashes extracted from Windows SAM-related sources.

Initial testing against a shadow file did not produce usable results. After switching to SAMdump2-derived data, Ophcrack successfully recovered 5 of 12 passwords in approximately 1 hour and 25 minutes. This indicates that Ophcrack is effective when the source format is compatible with the tool's expected workflow, but less useful when the wrong credential source is provided.

2. Johnny and John the Ripper

Johnny, the graphical interface for John the Ripper, and the John the Ripper command-line utility were used to evaluate broader password recovery flexibility. The GUI workflow successfully recovered 7 passwords from Windows-oriented data.

John the Ripper provided stronger input flexibility than Johnny because it supported both SAMPWdump-derived files and shadow-file based input. In this assessment, John the Ripper recovered 7 passwords from SAMPWdump data and 6 passwords from the shadow-file workflow.

3. Hash Identification Workflow

A hash identification utility available in Kali Linux was used to determine the likely algorithm type of selected hashes before Hashcat execution. This reduced trial-and-error when selecting the correct mode for Hashcat and helped validate that the hashes being processed matched the expected recovery workflow.

4. Hashcat Troubleshooting and Recovery

Hashcat was evaluated as the final recovery tool because of its broad algorithm support and high-performance cracking capability. The assessment showed that Hashcat was effective, but only after several workflow problems were corrected.

The first problem was incorrect algorithm selection. An initial attempt used a mode intended for MD5 rather than the actual hash type. The second issue involved incorrect shadow-file extraction. An initial tail command copied only the final hash instead of the full set, and a later extraction still omitted the root hash. After the missing hash was reinserted and the file structure was corrected, Hashcat successfully recovered 6 hashes.

Recommendations

- Validate the source format before choosing a recovery tool. Windows SAM-derived data and Unix/Linux shadow-file data often require different workflows.
- Use a hash identification utility before running Hashcat or other algorithm-specific tools to reduce incorrect mode selection.
- Verify that extracted hash files are complete and ordered correctly before beginning recovery attempts, especially when using command-line filtering or extraction commands.
- Use John the Ripper when broader source compatibility is needed across both Windows and shadow-file style inputs.
- Treat recovered weak passwords as evidence of weak password hygiene and use the results to support stronger password policy, passphrases, and multi-factor authentication.

Conclusion

This assessment demonstrated that password recovery success depends on tool selection, input source compatibility, hash identification, and correct file preparation. Ophcrack performed reasonably well when compatible Windows hash data was supplied, while Johnny and John the Ripper provided stronger operational flexibility across multiple input types. Hashcat ultimately produced strong results after troubleshooting corrected algorithm and file-format issues.

Overall, the assessment reinforces the importance of strong password controls, careful credential handling, and disciplined preprocessing when conducting password auditing and hash recovery work.

Appendix A: Assessment Evidence

The following figures provide supporting technical evidence for the primary assessment activities, including password recovery results captured from Ophcrack, Johnny, John the Ripper, and the hash identification workflow.

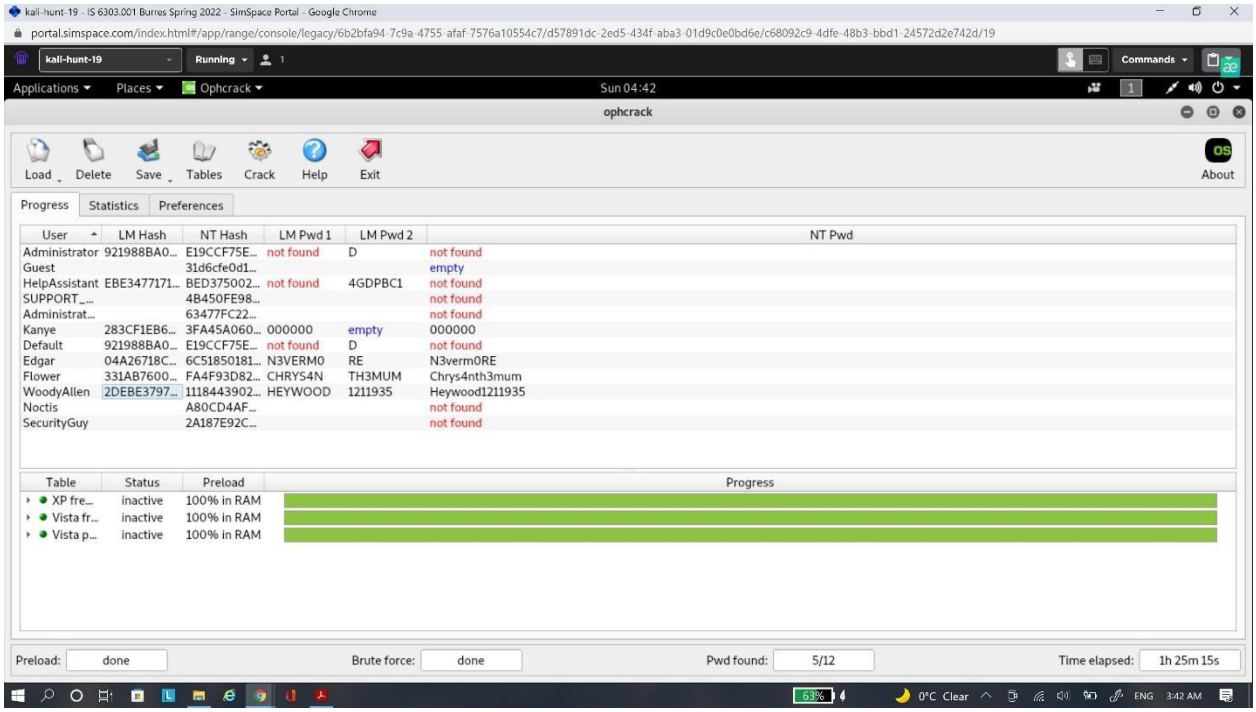

Figure A1. Ophcrack results showing Windows password recovery activity and five recovered passwords from twelve imported entries.

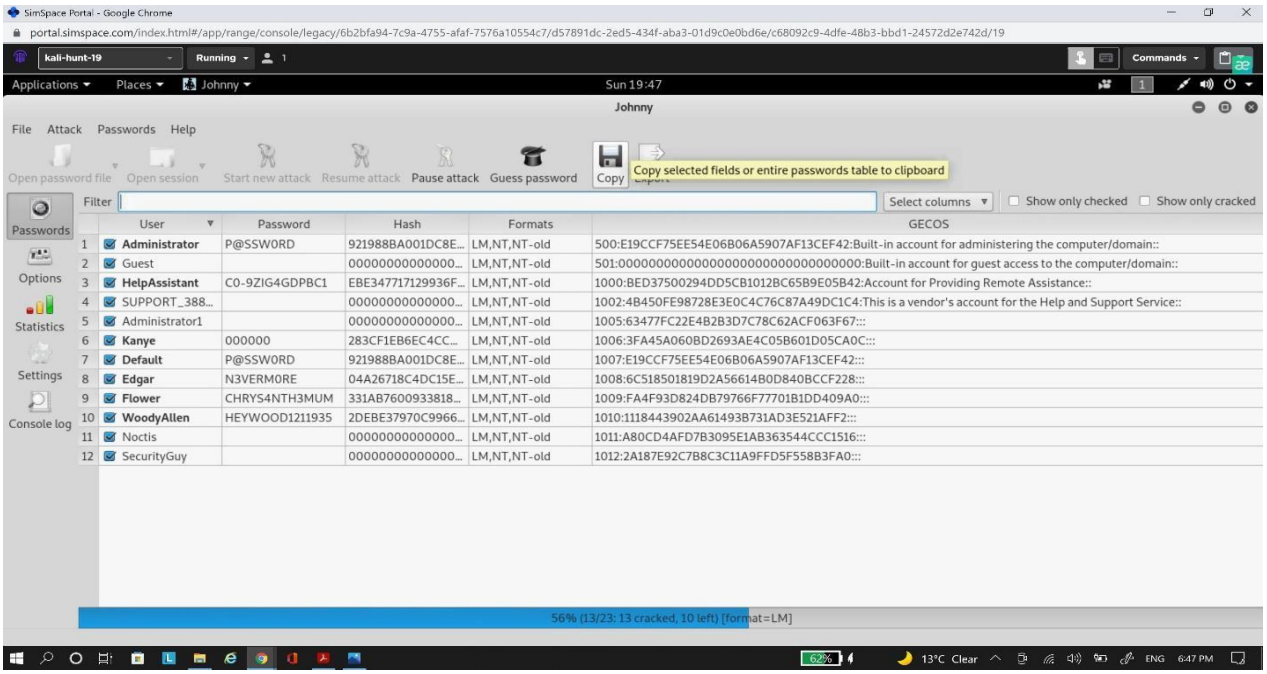

Figure A2. Johnny interface displaying recovered Windows passwords and corresponding hash entries during GUI-based analysis.

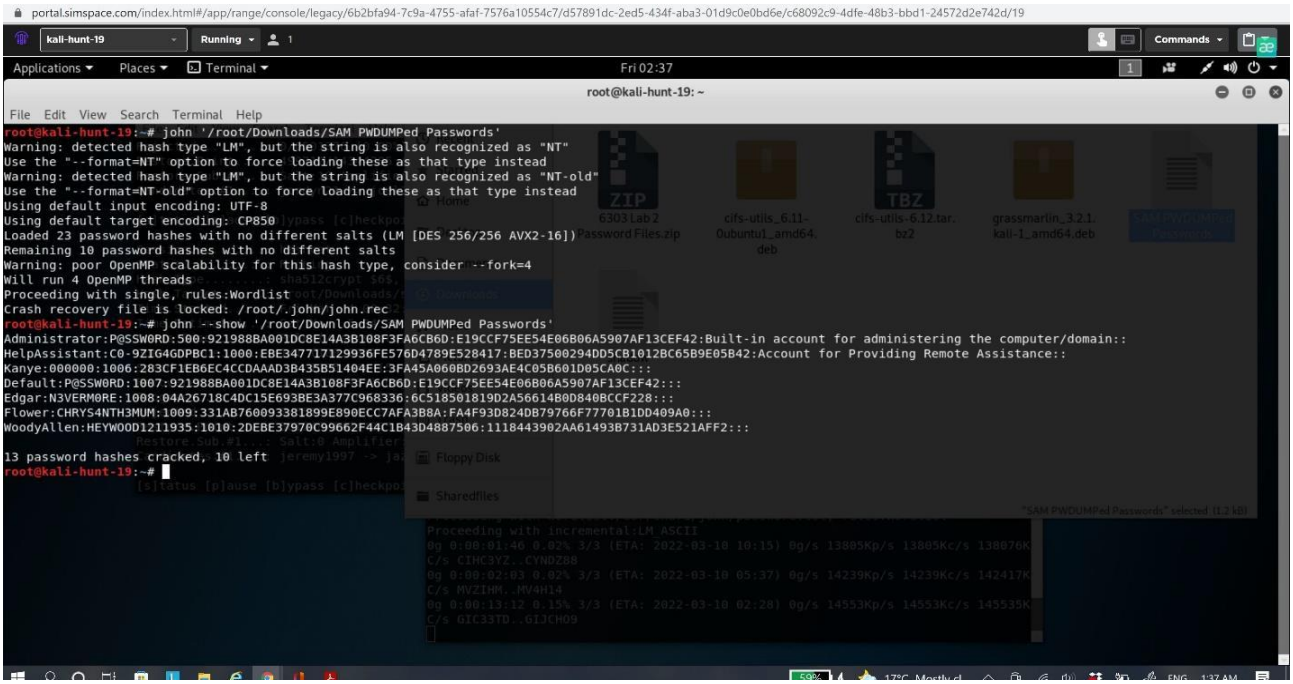

Figure A3. John the Ripper command-line output showing recovery results against SAMPWdump-derived password hashes.

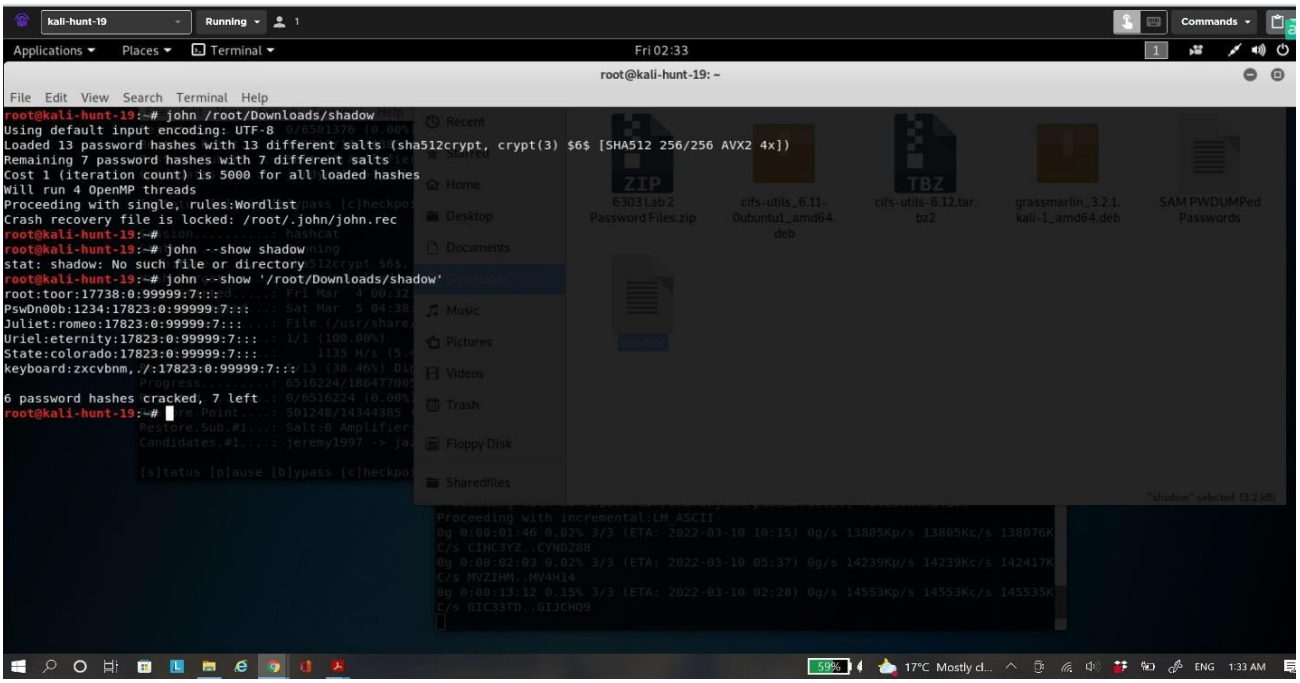

Figure A4. John the Ripper output showing recovered passwords from shadow-file based hash input.

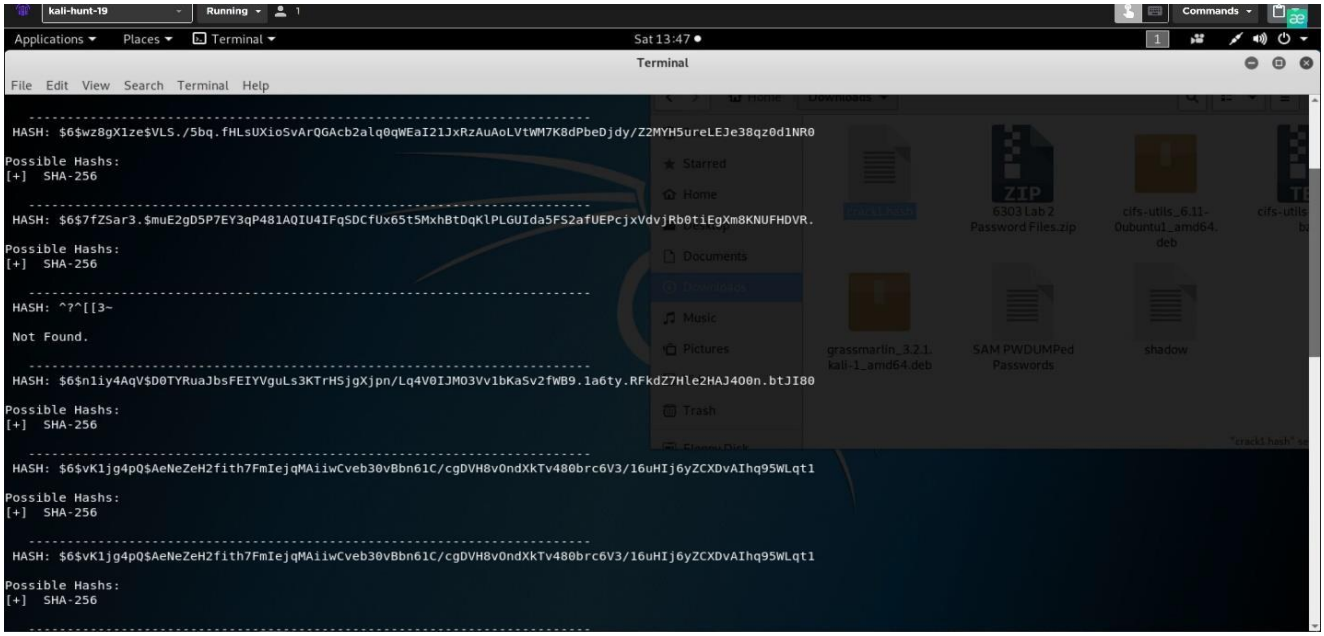

Figure A5. Hash identification output used to determine likely hash types prior to Hashcat execution.

Appendix B: Hashcat Troubleshooting and Final Recovery Evidence

These figures document the final Hashcat workflow, including algorithm selection, recovery execution, and the resulting cracked-hash output after troubleshooting file-format and mode-selection issues.

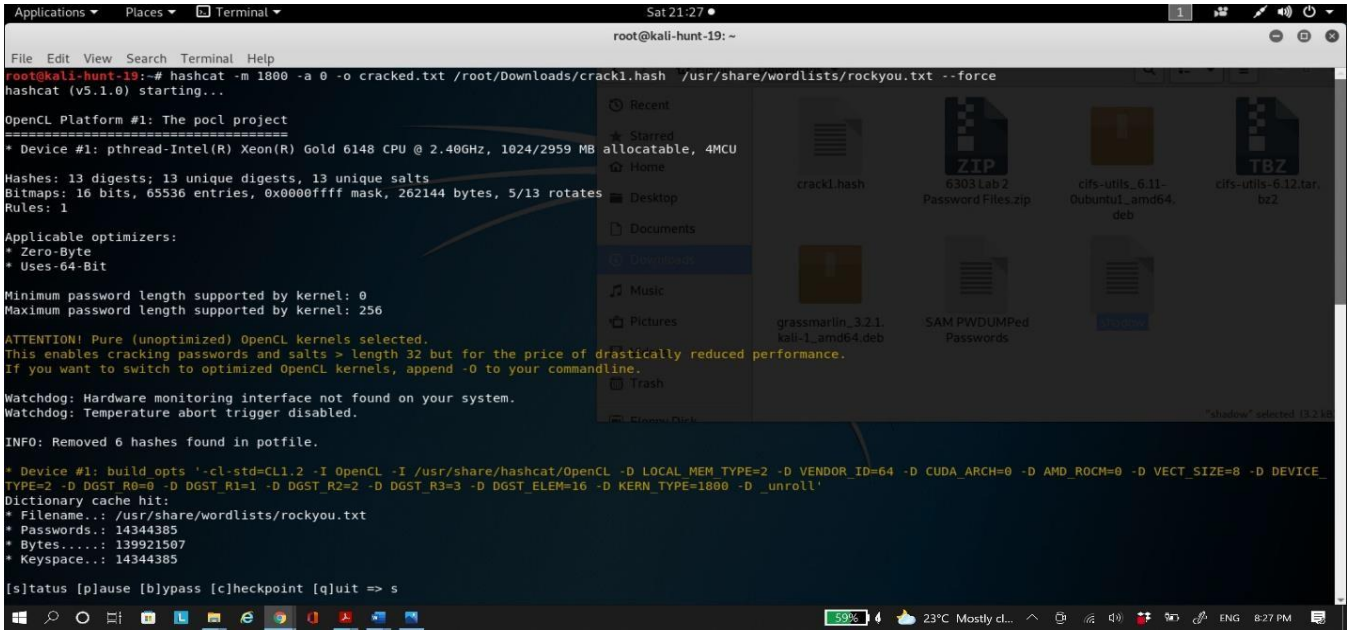

Figure B1. Hashcat execution showing the final algorithm selection and active cracking session after workflow corrections were applied.

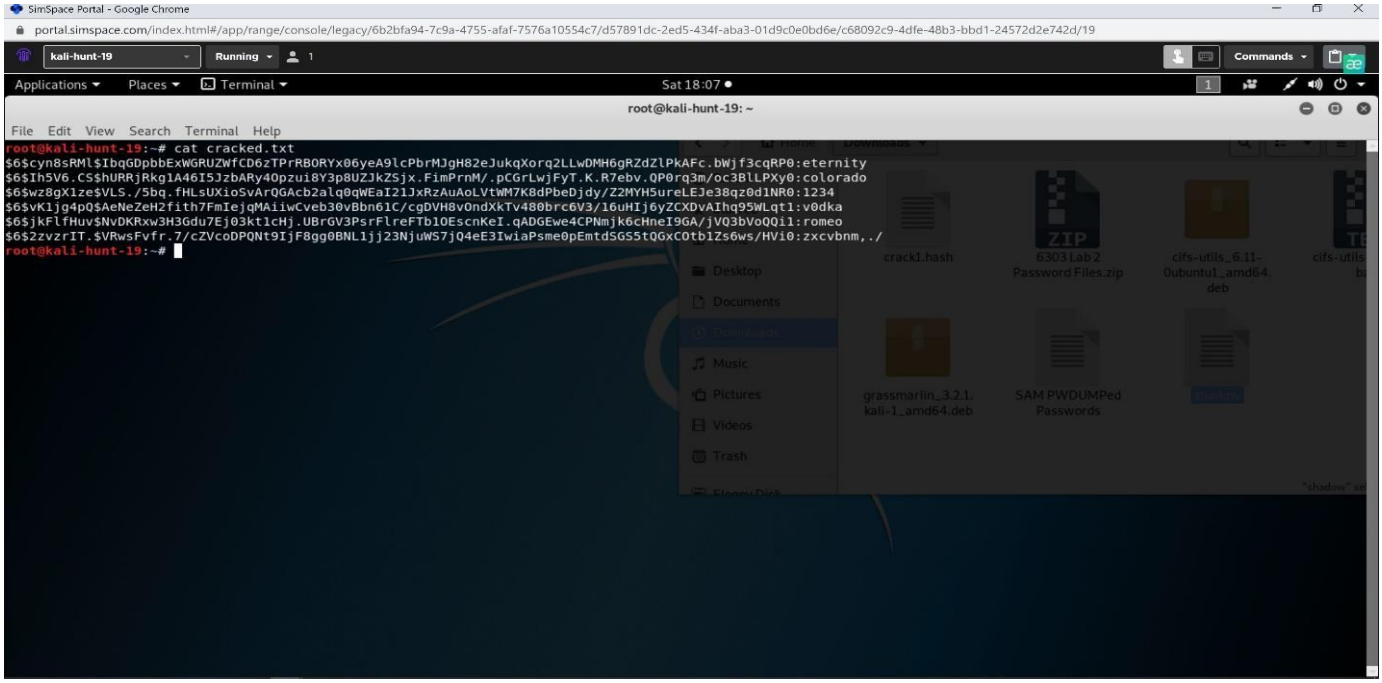

Figure B2. Final cracked-hash output from Hashcat showing six recovered hash values after repeated troubleshooting and reruns.